

MATH 152 — PART 3

Functions, Limits, and Continuity
Detailed Notes with Tips and Examples

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3.1 Introduction to Functions

Definition of a Function

A **function** f is a relation that takes elements from a set **D** (called the **domain**) to elements of another set (called the **codomain**) such that **each element in the domain has exactly one image** in the codomain.

🔑 **KEY INSIGHT:** A function is a "machine" that takes an input and produces exactly one output.

Input $(x) \rightarrow [f] \rightarrow$ Output $(f(x))$

Notation:

$f: x \rightarrow f(x)$ or $y = f(x)$

- x** = independent variable (input)
- y** = dependent variable (output)
- Domain** = all possible x values
- Range** = all possible y values (outputs)

Domain of a Function

The domain is the set of all real numbers for which the function is **defined**.

Common Domain Restrictions:

Function Type	Restriction	Example
Rational	Denominator $\neq 0$	$f(x) = 1/x \rightarrow x \neq 0$
Square root	Radicand ≥ 0	$f(x) = \sqrt{x} \rightarrow x \geq 0$
Even root	Radicand ≥ 0	$f(x) = \sqrt[4]{x} \rightarrow x \geq 0$
Odd root	No restriction	$f(x) = \sqrt[3]{x} \rightarrow$ all real
Logarithm	Argument > 0	$f(x) = \ln(x) \rightarrow x > 0$
Tangent	$x \neq \pi/2 + n\pi$	$f(x) = \tan x$

💡 **TIP:** When finding domain, always ask: "What values of x would make this undefined?"

Examples of Finding Domain

Example 1: $f(x) = 1/(x - 3)$

Domain: $x - 3 \neq 0 \rightarrow x \neq 3$
Domain = $\mathbb{R} \setminus \{3\} = (-\infty, 3) \cup (3, \infty)$

Example 2: $f(x) = \sqrt{x + 5}$

Domain: $x + 5 \geq 0 \rightarrow x \geq -5$
Domain = $[-5, \infty)$

Example 3: $f(x) = \sqrt{x^2 - 4}$

Domain: $x^2 - 4 \geq 0 \rightarrow (x - 2)(x + 2) \geq 0$
Solution: $x \leq -2$ or $x \geq 2$
Domain = $(-\infty, -2] \cup [2, \infty)$

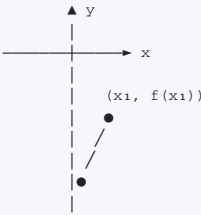
Example 4: $f(x) = \ln(x^2 - 1)$

Domain: $x^2 - 1 > 0 \rightarrow (x - 1)(x + 1) > 0$
Solution: $x < -1$ or $x > 1$
Domain = $(-\infty, -1) \cup (1, \infty)$

Graph of a Function

The graph of $y = f(x)$ is a pictorial representation of the function on a rectangular coordinate system.

➡️**KEY INSIGHT:** For a relation to be a function, no vertical line can intersect the graph more than once (Vertical Line Test).



💡**TIP:** The vertical line test is a quick way to check if a graph represents a function.

3.2 Types of Functions

1. Polynomial Functions

Definition: Functions of the form:

$$P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n, \text{ where } a_n \neq 0$$

- **n** = degree of the polynomial
- **a₀, a₁, ..., a_n** = constants (coefficients)
- **n** = positive integer

Function	Degree	Type
$f(x) = 3x + 2$	1	Linear
$f(x) = x^2 - 4x + 3$	2	Quadratic
$f(x) = 2x^3 - x + 5$	3	Cubic
$f(x) = x^4 - 2x^2 + 1$	4	Quartic

Important Theorem:

- Every polynomial equation $f(x) = 0$ has at least one root (Fundamental Theorem of Algebra)
- If degree is n, there are exactly n roots (counting multiplicity)

2. Algebraic Functions

Definition: Any function $y = f(x)$ satisfying an equation:

$$P_0(x)y^n + P_1(x)y^{n-1} + \dots + P_n(x) = 0$$

where $P_0, P_1, \dots, P_n$ are polynomials in x .

Type	Description	Example
Rational	$P(x)/Q(x)$ where P,Q are polynomials	$f(x) = 1/x^3$
Irrational	Cannot be expressed as quotient of polynomials	$f(x) = \sqrt{x}$

Example: $f(x) = x/(x+1)$ is algebraic since: $(x+1) \cdot f(x) - x = 0$

3. Transcendental Functions

Definition: Any function that is **not algebraic** is transcendental.

Key Characteristics:

- Cannot be expressed as a finite polynomial
- Usually expressed as infinite sums
- Examples include trigonometric, exponential, and logarithmic functions

4. Elementary Transcendental Functions

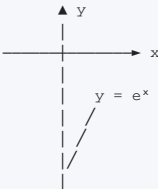
(a) Exponential Functions

$$f(x) = a^x, \text{ where } a > 0, a \neq 1$$

Properties:

- Always positive ($a^x > 0$)
- Passes through (0, 1)
- If $a > 1$: increasing function
- If $0 < a < 1$: decreasing function

Special Case: e^x where $e \approx 2.71828$ (natural base)



(b) Logarithmic Functions

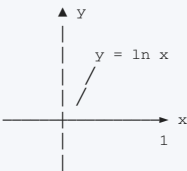
$$f(x) = \log_a x, \text{ where } a > 0, a \neq 1$$

Properties:

- Domain: $(0, \infty)$
- Range: $(-\infty, \infty)$
- Passes through (1, 0)
- If $a > 1$: increasing function
- If $0 < a < 1$: decreasing function

Special Cases: Common logarithm: $\log_{10} x$ (base 10). Natural logarithm: $\ln x = \log_e x$ (base e).

🔑**KEY INSIGHT:** $y = a^x$ and $y = \log_a x$ are inverse functions.



(c) Trigonometric Functions

Basic Functions:

$$\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$$

Fundamental Identities:

Identity	Name
$\sin^2x + \cos^2x = 1$	Pythagorean
$1 + \tan^2x = \sec^2x$	Pythagorean
$1 + \cot^2x = \csc^2x$	Pythagorean
$\tan x = \sin x/\cos x$	Quotient
$\cot x = \cos x/\sin x$	Quotient

Addition Formulas:

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$
$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$
$$\tan(x + y) = (\tan x + \tan y) / (1 - \tan x \tan y)$$

Double Angle Formulas:

$$\sin 2x = 2 \sin x \cos x$$
$$\cos 2x = \cos^2x - \sin^2x = 2\cos^2x - 1 = 1 - 2\sin^2x$$
$$\tan 2x = 2\tan x / (1 - \tan^2x)$$

(d) Inverse Trigonometric Functions

$$f(x) = \sin^{-1}x \quad (\arcsin x)$$
$$f(x) = \cos^{-1}x \quad (\arccos x)$$
$$f(x) = \tan^{-1}x \quad (\arctan x)$$

Function	Domain	Range
arcsin x	[−1, 1]	[−π/2, π/2]
arccos x	[−1, 1]	[0, π]
arctan x	(−∞, ∞)	(−π/2, π/2)

⚠**CRITICAL WARNING:** $\sin^{-1}x$ means the INVERSE function, NOT $1/\sin x$!

(e) Hyperbolic Functions

Definitions:

$$\sinh x = (e^x - e^{-x}) / 2$$
$$\cosh x = (e^x + e^{-x}) / 2$$
$$\tanh x = \sinh x / \cosh x = (e^x - e^{-x}) / (e^x + e^{-x})$$
$$\operatorname{csch} x = 1 / \sinh x = 2 / (e^x - e^{-x})$$
$$\operatorname{sech} x = 1 / \cosh x = 2 / (e^x + e^{-x})$$
$$\operatorname{coth} x = 1 / \tanh x = (e^x + e^{-x}) / (e^x - e^{-x})$$

Properties:

- $\cosh^2x - \sinh^2x = 1$
- $\sinh(-x) = -\sinh x$ (odd)
- $\cosh(-x) = \cosh x$ (even)

3.3 Bounded Functions

Definition

A function $f(x)$ is **bounded above** on $[a,b]$ if:

$$\exists M \text{ such that } f(x) \leq M, \forall x \in [a,b]$$

A function $f(x)$ is **bounded below** on $[a,b]$ if:

$$\exists m \text{ such that } f(x) \geq m, \forall x \in [a,b]$$

A function is **bounded** if it is both bounded above and below:

$$m \leq f(x) \leq M, \forall x \in [a,b]$$

Examples

Example 1: $f(x) = 4 - x$ on $[-1, 1]$

Upper bound: 5 (or any larger)
Lower bound: 3 (or any smaller)
Bounded? Yes

Example 2: $f(x) = 1/x$ on $(0, 1)$

Upper bound: None (goes to ∞ as $x \rightarrow 0^+$)
Lower bound: 1 (when $x = 1$)
Bounded? No (not bounded above)

Example 3: $f(x) = \sin x$ on $\mathbb{R}$

$-1 \leq \sin x \leq 1$
Upper bound: 1
Lower bound: -1
Bounded? Yes

3.4 Monotonic Functions

Definition

Type	Condition
Monotonic increasing	$x_1 < x_2 \rightarrow f(x_1) \leq f(x_2)$
Strictly increasing	$x_1 < x_2 \rightarrow f(x_1) < f(x_2)$
Monotonic decreasing	$x_1 < x_2 \rightarrow f(x_1) \geq f(x_2)$
Strictly decreasing	$x_1 < x_2 \rightarrow f(x_1) > f(x_2)$

Examples

Example 1: $f(x) = 2x + 1$ on $[0, 1]$

If $x_1 < x_2$, then $2x_1 + 1 < 2x_2 + 1$
Therefore, $f(x)$ is strictly increasing.

Example 2: $f(x) = 1/x$ on $[1, 3]$

If $x_1 < x_2$, then $1/x_1 > 1/x_2$
Therefore, $f(x)$ is strictly decreasing.

Example 3: $f(x) = (x+1)/x$ on $[1, 5]$

$f(x) = 1 + 1/x$
If $x_1 < x_2$, then $1/x_1 > 1/x_2$
Therefore, $f(x)$ is strictly decreasing.

3.5 Maxima and Minima

Definitions

Absolute Maximum: $f(x_0)$ is an absolute maximum on $[a,b]$ if:

$$f(x) \leq f(x_0), \quad \forall x \in [a,b]$$

Absolute Minimum: $f(x_0)$ is an absolute minimum on $[a,b]$ if:

$$f(x) \geq f(x_0), \quad \forall x \in [a,b]$$

Relative Maximum: $f(x_0)$ is a relative maximum if the above holds only in some deleted δ -neighbourhood of x_0 .

Relative Minimum: $f(x_0)$ is a relative minimum if the above holds only in some deleted δ -neighbourhood of x_0 .

Example

$y = x^2 \text{ on } [-2, 2]$

Absolute minimum: $f(0) = 0$ (lowest point)
Absolute maximum: $f(2) = f(-2) = 4$ (highest points)

3.6 Even and Odd Functions

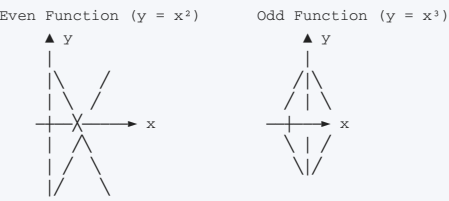
Definitions

Even Function: $f(-x) = f(x)$ for all x

Odd Function: $f(-x) = -f(x)$ for all x

Symmetry:

- Even functions are symmetric about the y-axis
- Odd functions have rotational symmetry of order 2 about the origin



Examples

Example 1: $f(x) = x^2$

$$f(-x) = (-x)^2 = x^2 = f(x)$$

Therefore, $f(x)$ is EVEN.

Example 2: $f(x) = x^3 - 2x$

$$f(-x) = (-x)^3 - 2(-x) = -x^3 + 2x = -(x^3 - 2x) = -f(x)$$

Therefore, $f(x)$ is ODD.

Example 3: $f(x) = x^2 + x$

$$f(-x) = (-x)^2 + (-x) = x^2 - x$$

Neither $f(x)$ nor $-f(x)$

Therefore, $f(x)$ is NEITHER even nor odd.

Example 4: $f(x) = \sin x$

$$\sin(-x) = -\sin x$$

Therefore, $\sin x$ is ODD.

Example 5: $f(x) = \cos x$

$$\cos(-x) = \cos x$$

Therefore, $\cos x$ is EVEN.

Decomposition into Even and Odd Parts

Any function $f(x)$ can be expressed as the sum of an even and odd function:

$$f(x) = \underbrace{\frac{1}{2}[f(x) + f(-x)]}_{\text{Even part}} + \underbrace{\frac{1}{2}[f(x) - f(-x)]}_{\text{Odd part}}$$

Example: $f(x) = (x+1)^2$

$$f(-x) = (-x+1)^2 = (1-x)^2$$

Even part = $\frac{1}{2}[(x+1)^2 + (1-x)^2]$

Odd part = $\frac{1}{2}[(x+1)^2 - (1-x)^2]$

3.7 Periodic Functions

Definition

A function $f(x)$ is **periodic** if $\exists$ a real number T such that:

$f(x + T) = f(x), \text{ for all } x$

T is called the **period** (usually the smallest positive such number).

Examples

Example 1: $f(x) = \sin x$

$\sin(x + 2\pi) = \sin x$
Period = 2π

Example 2: $f(x) = \tan x$

$\tan(x + \pi) = \tan x$
Period = π

Example 3: $f(x) = \sin(2x)$

$\sin(2(x + \pi)) = \sin(2x + 2\pi) = \sin(2x)$
Period = π

Example 4: $f(x) = |x|$

Not periodic (no $T > 0$ such that $|x+T| = |x|$ for all x)

3.8 Limits of Functions

Intuitive Definition

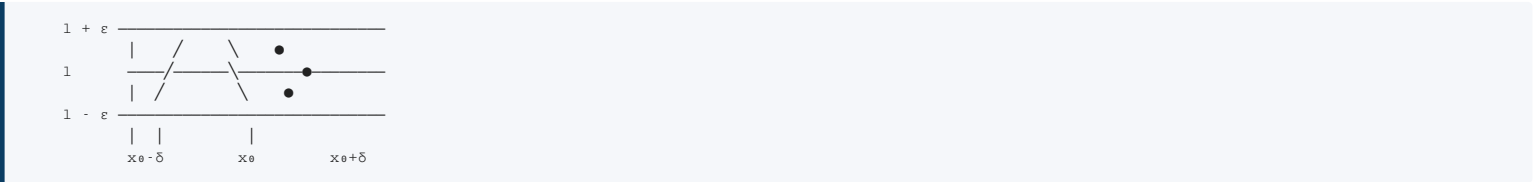
$f(x)$ has limit l as x approaches x_0 if $f(x)$ can be made **as close as we please** to l by choosing x **sufficiently close** to x_0 .

$$\lim_{(x \rightarrow x_0)} f(x) = l$$

Formal ϵ - δ Definition

$\lim_{(x \rightarrow x_0)} f(x) = l$ if for **every** $\epsilon > 0$, $\exists \delta > 0$ such that:

$$|f(x) - l| < \epsilon, \text{ whenever } 0 < |x - x_0| < \delta$$



One-Sided Limits

Left-Hand Limit (LHL)

$$\lim_{(x \rightarrow x_0^-)} f(x) = l_1$$

Approaches x_0 from the LEFT ($x < x_0$)

Right-Hand Limit (RHL)

$$\lim_{(x \rightarrow x_0^+)} f(x) = l_2$$

Approaches x_0 from the RIGHT ($x > x_0$)

➡ **KEY INSIGHT:** Two-sided limit exists when $l_1 = l_2 = l$. Then: $\lim_{(x \rightarrow x_0)} f(x) = l$

Examples of One-Sided Limits

Example 1: $f(x) = 2x + 1$

$$\lim_{(x \rightarrow 0^-)} (2x+1) = 1$$
$$\lim_{(x \rightarrow 0^+)} (2x+1) = 1$$

Both equal $\rightarrow$ two-sided limit exists.

Example 2: $f(x) = 1/x$

$$\lim_{(x \rightarrow 0^-)} (1/x) = -\infty$$
$$\lim_{(x \rightarrow 0^+)} (1/x) = +\infty$$

Different $\rightarrow$ two-sided limit does NOT exist.

Example 3: $f(x) = \tan^{-1}(1/x)$

$$\lim_{(x \rightarrow 0^-)} \tan^{-1}(1/x) = -\pi/2$$
$$\lim_{(x \rightarrow 0^+)} \tan^{-1}(1/x) = \pi/2$$

Different $\rightarrow$ two-sided limit does NOT exist.

Example 4: $f(x) = |x-1|/(x-1)$

For $x < 1$: $|x-1| = -(x-1) \rightarrow f(x) = -1$
For $x > 1$: $|x-1| = (x-1) \rightarrow f(x) = 1$

$$\lim_{(x \rightarrow 1^-)} f(x) = -1$$
$$\lim_{(x \rightarrow 1^+)} f(x) = 1$$

Different $\rightarrow$ two-sided limit does NOT exist.

Piecewise Functions

EXAMPLE

$$f(x) = \begin{cases} -1, & \text{if } x < 1 \\ x, & \text{if } -1 \leq x \leq 1 \\ 1, & \text{if } x > 1 \end{cases}$$
$$\lim_{x \rightarrow -1^-} f(x) = -1$$
$$\lim_{x \rightarrow -1^+} f(x) = -1 \quad (\text{since } f(x) = x \text{ at } x = -1)$$

Therefore, $\lim_{x \rightarrow -1} f(x) = -1$ EXISTS.

$$\lim_{x \rightarrow 1^-} f(x) = 1 \quad (\text{since } f(x) = x \text{ at } x = 1)$$
$$\lim_{x \rightarrow 1^+} f(x) = 1$$

Therefore, $\lim_{x \rightarrow 1} f(x) = 1$ EXISTS.

3.9 Theorems on Limits

Algebra of Limits

If $\lim_{x \rightarrow x_0} f(x) = A$ and $\lim_{x \rightarrow x_0} g(x) = B$, then:

Operation	Formula
Sum	$\lim(f + g) = A + B$
Difference	$\lim(f - g) = A - B$
Product	$\lim(f \cdot g) = A \cdot B$
Quotient	$\lim(f/g) = A/B \text{ (} B \neq 0 \text{)}$
Power	$\lim(f^p) = A^p$

Special Cases for Quotient:

- If $B = 0$ and $A \neq 0$: limit does NOT exist
- If $B = 0$ and $A = 0$: indeterminate form (may or may not exist)

Evaluating Limits

Rule 1: Polynomials

If $f(x)$ is a polynomial:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

Example: $\lim_{x \rightarrow 1} (x^2 + 2x - 5) = 1 + 2 - 5 = -2$

Rule 2: Elementary Functions

If $f(x)$ is an elementary function:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

Example: $\lim_{x \rightarrow 0} e^x = e^0 = 1$ $\lim_{x \rightarrow \pi/2} \sin x = \sin(\pi/2) = 1$

Rule 3: Common Factor

If $f(x) = (x - x_0)^a \cdot g(x) / (x - x_0)^a \cdot h(x)$:

$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) / h(x)$$

EXAMPLE

$$\begin{aligned} &\lim_{x \rightarrow 2} (x^3 - 8) / (x - 2) \\ &= \lim_{x \rightarrow 2} (x - 2)(x^2 + 2x + 4) / (x - 2) \\ &= \lim_{x \rightarrow 2} (x^2 + 2x + 4) \\ &= 4 + 4 + 4 = 12 \end{aligned}$$

Important Limits to Memorize

Limit	Value
$\lim_{x \rightarrow 0} \sin x / x$	1
$\lim_{x \rightarrow 0} (1 - \cos x) / x^2$	1/2
$\lim_{x \rightarrow \infty} (1 + 1/x)^x$	e
$\lim_{x \rightarrow 0^+} (1 + x)^{1/x}$	e
$\lim_{x \rightarrow 0} (e^x - 1) / x$	1
$\lim_{x \rightarrow 0} \ln(1 + x) / x$	1

Limits at Infinity

Definition: $\lim_{x \rightarrow \infty} f(x) = L$ if for every $\varepsilon > 0$, $\exists M$ such that:

$$|f(x) - L| < \varepsilon \text{ whenever } x > M$$

Examples:

$$\begin{aligned} \lim_{x \rightarrow \infty} 1/x &= 0 \\ \lim_{x \rightarrow \infty} (3x^2 + 7x - 6) / (4x^2 + 3x + 6) &= 3/4 \end{aligned}$$

Method: Divide numerator and denominator by highest power of x.

EXAMPLE

$$\lim_{x \rightarrow \infty} \frac{(3x^2+7x-6)}{(4x^2+3x+6)}$$

Divide by x^2 :

$$= \lim_{x \rightarrow \infty} \frac{(3 + 7/x - 6/x^2)}{(4 + 3/x + 6/x^2)}$$
$$= \frac{(3 + 0 - 0)}{(4 + 0 + 0)}$$
$$= 3/4$$

Indeterminate Forms

Form	Example	Solution Method
0/0	$\lim_{x \rightarrow 0} \sin x/x$	Factor, L'Hospital, or known limit
∞/∞	$\lim_{x \rightarrow \infty} (x^2+1)/(2x^2+3)$	Divide by highest power
$0 \times \infty$	$\lim_{x \rightarrow 0^+} x \cdot \ln x$	Rewrite as fraction
$\infty - \infty$	$\lim_{x \rightarrow 0} (1/x - 1/\sin x)$	Combine terms
1^∞	$\lim_{x \rightarrow \infty} (1+1/x)^x$	Use e formula

3.10 L'Hospital's Rule

Statement

If $\lim_{x \rightarrow c} f(x)/g(x)$ gives an indeterminate form (0/0 or ∞/∞), then:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided the limit of the derivatives exists.

Examples

EXAMPLE 1

$$\lim_{x \rightarrow 1} \frac{(x^3 + x^2 - x - 1)}{(x^2 + 2x - 3)}$$

Step 1: Check at x=1
Numerator: $1+1-1-1 = 0$
Denominator: $1+2-3 = 0$
Indeterminate (0/0)

Step 2: Apply L'Hospital
 $= \lim_{x \rightarrow 1} \frac{(3x^2 + 2x - 1)}{(2x + 2)}$
 $= (3+2-1)/(2+2) = 4/4 = 1$

EXAMPLE 2

$$\lim_{x \rightarrow 0} (1/x - 1/\sin x)$$

Step 1: Combine into single fraction
 $= \lim_{x \rightarrow 0} \frac{(\sin x - x)}{(x \sin x)}$

Step 2: Check at x=0
Numerator: $0-0 = 0$
Denominator: 0
Indeterminate (0/0)

Step 3: Apply L'Hospital
 $= \lim_{x \rightarrow 0} \frac{(\cos x - 1)}{(\sin x + x \cos x)}$

Step 4: Check again (still 0/0)
Apply L'Hospital again:
 $= \lim_{x \rightarrow 0} \frac{(-\sin x)}{(2\cos x - x \sin x)}$
 $= 0/(2 - 0) = 0$

EXAMPLE 3

$$\lim_{x \rightarrow 0^+} x \cdot \ln x$$

Step 1: Rewrite as fraction
 $= \lim_{x \rightarrow 0^+} \ln x / (1/x)$

Step 2: Check at x=0+
 $\ln x \rightarrow -\infty, 1/x \rightarrow \infty$
Indeterminate (∞/∞)

Step 3: Apply L'Hospital
 $= \lim_{x \rightarrow 0^+} (1/x) / (-1/x^2)$
 $= \lim_{x \rightarrow 0^+} -x$
 $= 0$

EXAMPLE 4

$$\lim_{x \rightarrow 0} \frac{(\sin x - x)}{(\tan x - x)}$$

Step 1: Check at x=0
 $0/0 \rightarrow$ indeterminate

Step 2: Apply L'Hospital (apply repeatedly until determined)
First: $\lim_{x \rightarrow 0} \frac{(\cos x - 1)}{(\sec^2 x - 1)} = 0/0$
Second: $\lim_{x \rightarrow 0} \frac{(-\sin x)}{(2\sec^2 x \cdot \tan x)} = 0/0$
Third: $\lim_{x \rightarrow 0} \frac{(-\cos x)}{(2\sec^2 x \cdot \sec^2 x + \dots)}$
 $= (-1)/(2) = -1/2$

When NOT to Use L'Hospital

⚠️**CRITICAL WARNING:** L'Hospital's Rule only works for $0/0$ or ∞/∞ forms.

Example: $\lim_{x \rightarrow 0} (1 + 2x)^{1/x}$

This is a 1^∞ form.
Don't use L'Hospital directly!
Use the formula: $\lim_{x \rightarrow 0} (1 + kx)^{1/x} = e^k$

 $= e^2$

3.11 Continuity

Definition of Continuity

A function $f(x)$ is **continuous at x_0** if:

- $\lim_{x \rightarrow x_0} f(x)$ exists**
- $f(x_0)$ is defined**
- $\lim_{x \rightarrow x_0} f(x) = f(x_0)$**

🔑**KEY INSIGHT:** A continuous function has no breaks, jumps, or holes at that point.

Types of Discontinuity

Type	Description	Example
Removable	Limit exists, but $f(x_0) \neq$ limit or undefined	$f(x) = (x^2-1)/(x-1)$ at $x=1$
Jump	Left and right limits exist but are different	$f(x) = x /x$ at $x=0$
Infinite	Function goes to $\pm\infty$	$f(x) = 1/x$ at $x=0$
Oscillating	No limit exists	$f(x) = \sin(1/x)$ at $x=0$

Examples

Example 1: $f(x) = (x^2 - 8)/(x - 3)$ at $x = 3$

$$f(3) \text{ is not defined}$$

Therefore, f is NOT continuous at $x = 3$.

Example 2

$$f(x) = \begin{cases} 1/x, & \text{if } x \neq 0 \\ 2, & \text{if } x = 0 \end{cases}$$

At $x = 0$:

$$\lim_{x \rightarrow 0} f(x) \text{ does not exist}$$

Therefore, f is NOT continuous at $x = 0$.

Example 3

$$f(x) = \begin{cases} (x^2 - 9)/(x - 3), & \text{if } x \neq 3 \\ 7, & \text{if } x = 3 \end{cases}$$

At $x = 3$:

$$\begin{aligned} \lim_{x \rightarrow 3} f(x) &= 6 \\ f(3) &= 7 \\ \lim &\neq f(3) \end{aligned}$$

Therefore, f is NOT continuous at $x = 3$.

Example 4

$$f(x) = \begin{cases} (x^2 - 9)/(x - 3), & \text{if } x \neq 3 \\ 6, & \text{if } x = 3 \end{cases}$$

At $x = 3$:

$$\begin{aligned} \lim_{x \rightarrow 3} f(x) &= 6 \\ f(3) &= 6 \\ \lim &= f(3) \end{aligned}$$

Therefore, f IS continuous at $x = 3$.

Right and Left Continuity

Type	Condition
Left continuous	$\lim_{x \rightarrow x_0^-} f(x) = f(x_0)$
Right continuous	$\lim_{x \rightarrow x_0^+} f(x) = f(x_0)$

Continuity in an Interval

Open Interval (a,b): f is continuous at every point $x_0 \in (a,b)$

Closed Interval [a,b]: f is continuous on (a,b); f is right continuous at a; f is left continuous at b

Theorems on Continuity

Theorem 1: Operations on Continuous Functions

If f and g are continuous at x_0 , then:

- $f \pm g$ is continuous at x_0
- $f \cdot g$ is continuous at x_0
- f/g is continuous at x_0 (provided $g(x_0) \neq 0$)

Theorem 2: Polynomials are Continuous

Every polynomial $P_n(x)$ is continuous for all real x.

Theorem 3: Composition of Continuous Functions

If f is continuous at x_0 and g is continuous at $f(x_0)$, then $g \circ f$ (composite function) is continuous at x_0 .

Theorem 4: Continuity Implies Boundedness

If f is continuous on a closed interval, then f is bounded on that interval.

Continuity and Differentiability Relationship

➡**KEY INSIGHT:** If f is differentiable at x_0 , then f is continuous at x_0 .

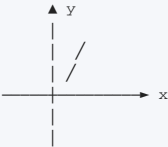
However: The converse is **NOT** true (differentiability $\Rightarrow$ continuity, but continuity $\nRightarrow$ differentiability).

COUNTEREXAMPLE — $f(x) = |x|$ at $x = 0$

Continuity: $\lim_{(x \rightarrow 0)} |x| = 0 = f(0) \rightarrow$ CONTINUOUS

Differentiability:
 $f'_{-}(0) = \lim_{(h \rightarrow 0^{-})} |h|/h = -1$
 $f'_{+}(0) = \lim_{(h \rightarrow 0^{+})} |h|/h = 1$
 $f'_{-}(0) \neq f'_{+}(0)$

Therefore, NOT DIFFERENTIABLE at $x = 0$.



3.12 Differentiability

Definition

The derivative of f at x_0 is:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Alternative Form:

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

One-Sided Derivatives

Type	Formula
Left derivative	$f'_-(x_0) = \lim_{h \rightarrow 0^-} \frac{f(x_0+h) - f(x_0)}{h}$
Right derivative	$f'_+(x_0) = \lim_{h \rightarrow 0^+} \frac{f(x_0+h) - f(x_0)}{h}$

🔑**KEY INSIGHT:** $f'(x_0)$ exists iff $f'_-(x_0) = f'_+(x_0)$

Examples

EXAMPLE 1 — $f(x) = x^2$ at $x = 2$

$$\begin{aligned} f'(2) &= \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} \\ &= \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\ &= \lim_{h \rightarrow 0} (4 + h) \\ &= 4 \end{aligned}$$

Therefore, f is differentiable at $x = 2$.

EXAMPLE 2

$$f(x) = \begin{cases} x, & \text{if } x \geq 1 \\ 2x-1, & \text{if } x < 1 \end{cases}$$

At $x = 1$:

$$\begin{aligned} f'_-(1) &= \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{(2(1+h)-1) - 1}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{2+2h-1-1}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{2h}{h} = 2 \end{aligned}$$
$$\begin{aligned} f'_+(1) &= \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{(1+h) - 1}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \end{aligned}$$

$f'_-(1) \neq f'_+(1)$
Therefore, f is NOT differentiable at $x = 1$.

3.13 Differentiation Rules

Basic Rules

Rule	Formula
Power Rule	$d/dx(x^n) = n \cdot x^{n-1}$
Sum Rule	$d/dx[f \pm g] = f' \pm g'$
Product Rule	$d/dx[f \cdot g] = f' \cdot g + f \cdot g'$
Quotient Rule	$d/dx[f/g] = (f' \cdot g - f \cdot g')/g^2$
Chain Rule	$d/dx[f(g(x))] = f'(g) \cdot g'$

Derivatives of Special Functions

Function	Derivative
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \cdot \tan x$
$\csc x$	$-\csc x \cdot \cot x$
e^x	e^x
a^x	$a^x \cdot \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \cdot \ln a)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$

PROBLEM 1

Is $f(x) = (x^2 - 4)/(x - 2)$ continuous at $x = 2$? If not, how can it be redefined to be continuous?

Solution:

```
f(2) = (4-4)/(2-2) = 0/0 (undefined)

lim(x→2) (x²-4)/(x-2) = lim(x→2) (x-2)(x+2)/(x-2)
                      = lim(x→2) (x+2) = 4

Since f(2) is undefined, f is NOT continuous.

To make it continuous, redefine:
f(x) = { (x²-4)/(x-2), if x ≠ 2
        { 4,           if x = 2
```

PROBLEM 2

Construct a function that is continuous at $x = 0$.

Solution:

Many possibilities! Here are a few:

```
1. f(x) = x (polynomials are always continuous)
2. f(x) = sin x (trig functions are continuous)
3. f(x) = e^x (exponential is continuous)
4. f(x) = { x, if x ≥ 0
           { 0, if x < 0
```

PROBLEM 3

Construct a function that is continuous at $x = 1$ but not differentiable there.

Solution:

```
f(x) = |x - 1|

At x = 1:
- Continuous: lim(x→1) |x-1| = 0 = f(1)
- Not differentiable:
  f'-(1) = -1, f'+(1) = 1
  Not equal → not differentiable
```

PROBLEM 4 — Differentiate the following

(a) $f(x) = (x^2+3)/(2x^3-1)$

Solution:

```
Using quotient rule:
f' = [(2x)(2x³-1) - (x²+3)(6x²)] / (2x³-1)²
    = [4x⁴ - 2x - 6x⁴ - 18x²] / (2x³-1)²
    = [-2x⁴ - 18x² - 2x] / (2x³-1)²
```

(b) $f(x) = \sin(x^2)$

Solution:

```
Using chain rule:
f'(x) = cos(x²) · 2x = 2x·cos(x²)
```

(c) $f(x) = \ln(3x + 1)$

Solution:

```
Using chain rule:
f'(x) = 1/(3x+1) · 3 = 3/(3x+1)
```

(d) $f(x) = e^{(2x+1)}$

Solution:

```
Using chain rule:
f'(x) = e^(2x+1) · 2 = 2e^(2x+1)
```

PROBLEM 5

Evaluate $\lim_{x \rightarrow -1} (x+2)(x^2+3)/(x^2-4)$

Solution:

Direct substitution:
 $x = -1$

$$\lim = (-1+2)(1+3)/(1-4) = (1)(4)/(-3) = -4/3$$

PROBLEM 6

Evaluate $\lim_{x \rightarrow 1} [1/(x-1)(1/(x+3) - 2/(3x+5))]$

Solution:

First simplify the inner bracket:
$$\begin{aligned} 1/(x+3) - 2/(3x+5) &= [(3x+5) - 2(x+3)] / [(x+3)(3x+5)] \\ &= [3x+5-2x-6] / [(x+3)(3x+5)] \\ &= [x-1] / [(x+3)(3x+5)] \end{aligned}$$

Now multiply by $1/(x-1)$:
$$\begin{aligned} &= (x-1) / [(x-1)(x+3)(3x+5)] \\ &= 1 / [(x+3)(3x+5)] \end{aligned}$$

Now take the limit:
$$\lim_{x \rightarrow 1} 1 / [(x+3)(3x+5)] = 1 / [(4)(8)] = 1/32$$

PROBLEM 7

Evaluate $\lim_{x \rightarrow \infty} (1+x)/(1-x)$

Solution:

Divide by x :
$$\begin{aligned} &= \lim_{x \rightarrow \infty} (1/x + 1)/(1/x - 1) \\ &= (0 + 1)/(0 - 1) = -1 \end{aligned}$$

PROBLEM 8

Evaluate $\lim_{x \rightarrow \infty} (x^3)/(x^3 + 3)$

Solution:

Divide by x^3 :
$$\begin{aligned} &= \lim_{x \rightarrow \infty} 1/(1 + 3/x^3) \\ &= 1/(1 + 0) = 1 \end{aligned}$$

PROBLEM 9

Evaluate $\lim_{x \rightarrow 0} (1 - \cos^2 x)/x^2$

Solution:

$$\begin{aligned} 1 - \cos^2 x &= \sin^2 x \\ &= \lim_{x \rightarrow 0} \sin^2 x / x^2 \\ &= \lim_{x \rightarrow 0} (\sin x / x)^2 \\ &= 1^2 = 1 \end{aligned}$$

PROBLEM 10

Evaluate $\lim_{x \rightarrow \pi/2} (\sec x - \tan x)$

Solution:

$$\sec x - \tan x = \frac{1}{\cos x} - \frac{\sin x}{\cos x} = \frac{1 - \sin x}{\cos x}$$

Check at $x = \pi/2$:

$$1 - \sin(\pi/2) = 1 - 1 = 0$$

$$\cos(\pi/2) = 0$$

Indeterminate $(0/0)$

Apply L'Hospital:

$$= \lim_{x \rightarrow \pi/2} \frac{(-\cos x)}{(-\sin x)}$$

$$= \lim_{x \rightarrow \pi/2} \cot x$$

$$= \cot(\pi/2) = 0$$

PROBLEM 11

Prove that $\lim_{x \rightarrow 0} \sin x / x = 1$

Solution:

We can use the squeeze theorem or known result.

$$\text{Let } f(x) = \sin x / x$$

For x close to 0 (but not 0):

$$\cos x < \sin x / x < 1$$

$$\lim_{x \rightarrow 0} \cos x = 1 \text{ and } \lim_{x \rightarrow 0} 1 = 1$$

By squeeze theorem:

$$\lim_{x \rightarrow 0} \sin x / x = 1$$

PROBLEM 12

Determine if $f(x) = \sin x / x$ is continuous at $x = 0$.

Solution:

$f(0)$ is undefined $(0/0)$

$$\lim_{x \rightarrow 0} \sin x / x = 1$$

To make it continuous:

$$f(x) = \begin{cases} \sin x / x, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$$

Common Mistakes and How to Avoid Them

Mistake 1: Forgetting Domain Restrictions

Wrong	Correct
$f(x) = 1/(x-2)$ has domain $\mathbb{R}$ ✗	$f(x) = 1/(x-2)$ has domain $\mathbb{R} \setminus \{2\}$ ✓

Mistake 2: Confusing Continuity with Differentiability

Wrong	Correct
$f(x)$ is continuous $\rightarrow f(x)$ is differentiable ✗	$f(x)$ is differentiable $\rightarrow f(x)$ is continuous ✓ (converse is false)

Mistake 3: Not Checking Both Sides of a Limit

Wrong	Correct
$\lim_{x \rightarrow 0} 1/x = 0$ ✗	$\lim_{x \rightarrow 0^-} 1/x = -\infty$ and $\lim_{x \rightarrow 0^+} 1/x = +\infty \rightarrow$ limit does NOT exist ✓

Mistake 4: Applying L'Hospital When Not Indeterminate

Wrong	Correct
$\lim_{x \rightarrow 1} (x^2+1)/(x+1) = \lim_{x \rightarrow 1} 2x/1 = 2$ ✗	At $x=1$: $(1+1)/(1+1) = 2/2 = 1$ — just substitute, not indeterminate! ✓

Mistake 5: Not Simplifying Before Finding Domain

⚠

CRITICAL WARNING: $f(x) = (x^2 - 4)/(x - 2) = (x-2)(x+2)/(x-2) = x+2$ — but the domain is STILL $x \neq 2$, despite the simplification!

Key Formulas Summary

Functions

Domain: Set of all valid inputs
Range: Set of all possible outputs
Vertical Line Test: Graph is a function if no vertical line hits twice

Even/Odd Functions

Even: $f(-x) = f(x)$
Odd: $f(-x) = -f(x)$

Limits

$\lim_{x \rightarrow x_0} f(x) = L$ if $|f(x) - L| < \epsilon$ when $0 < |x - x_0| < \delta$
 $\lim_{x \rightarrow x_0^-} f(x) = L_1$ (left)
 $\lim_{x \rightarrow x_0^+} f(x) = L_2$ (right)
Two-sided exists iff $L_1 = L_2$

Continuity

Three conditions:
1. $\lim$ exists
2. $f(x_0)$ defined
3. $\lim = f(x_0)$

Derivatives

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$